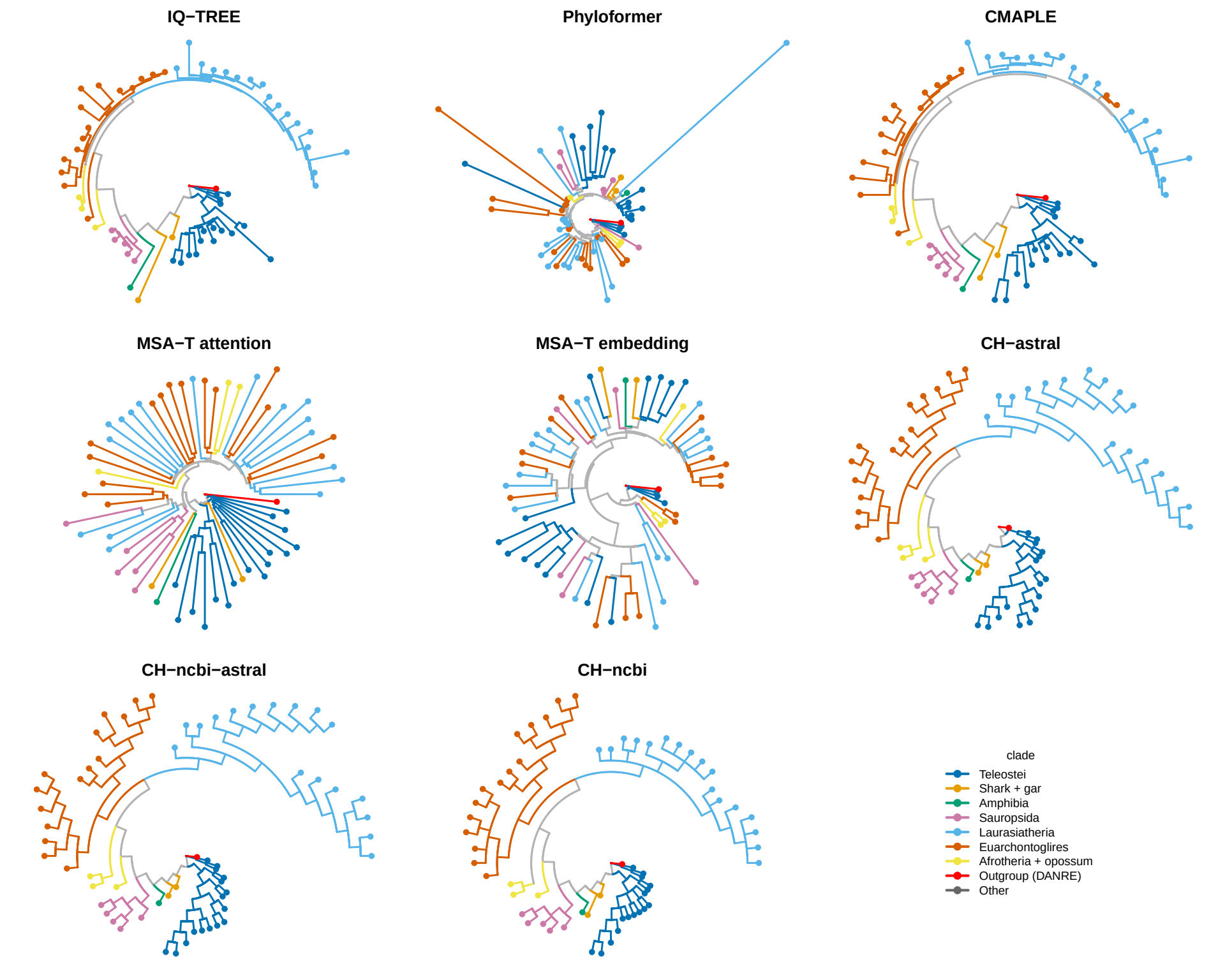
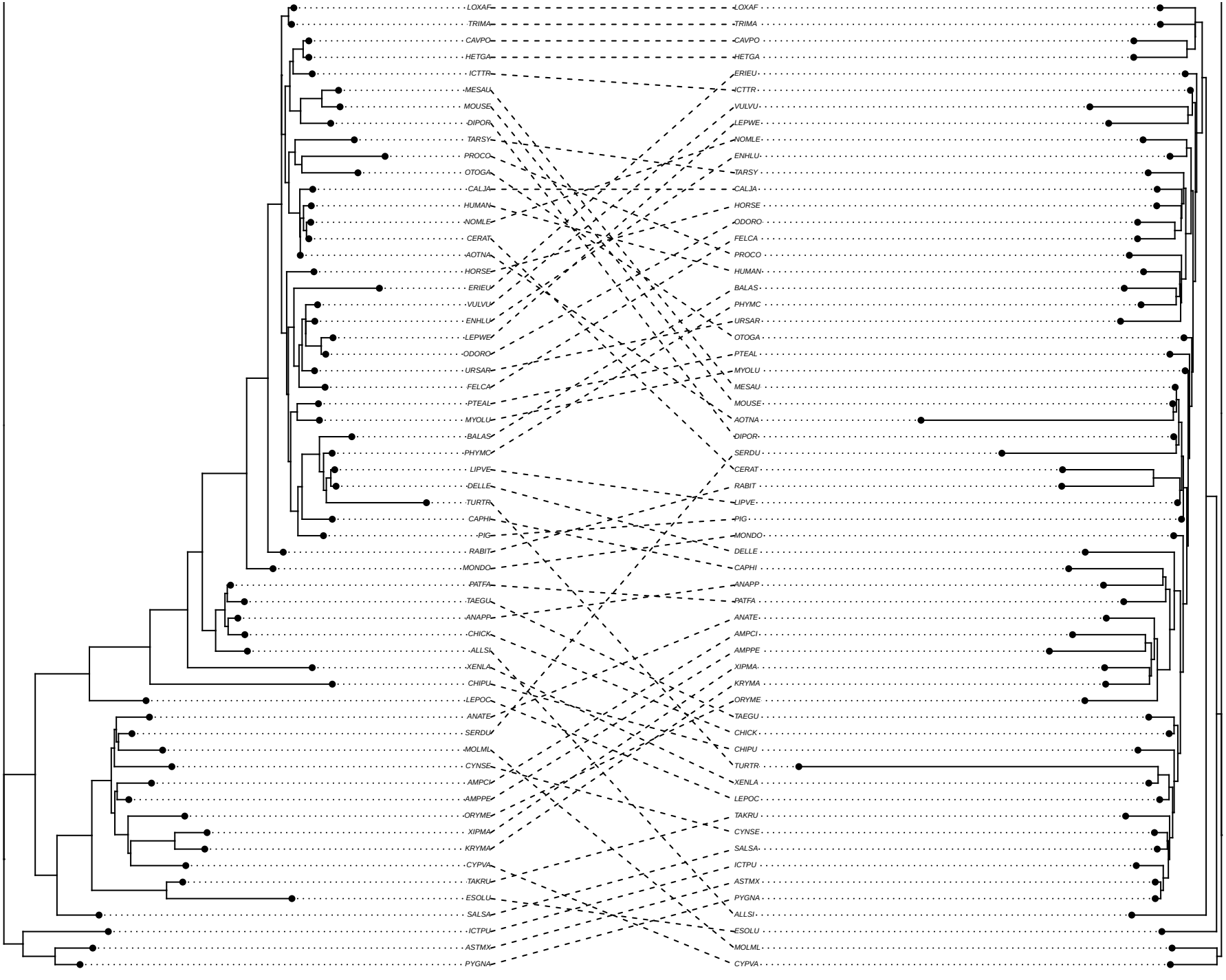
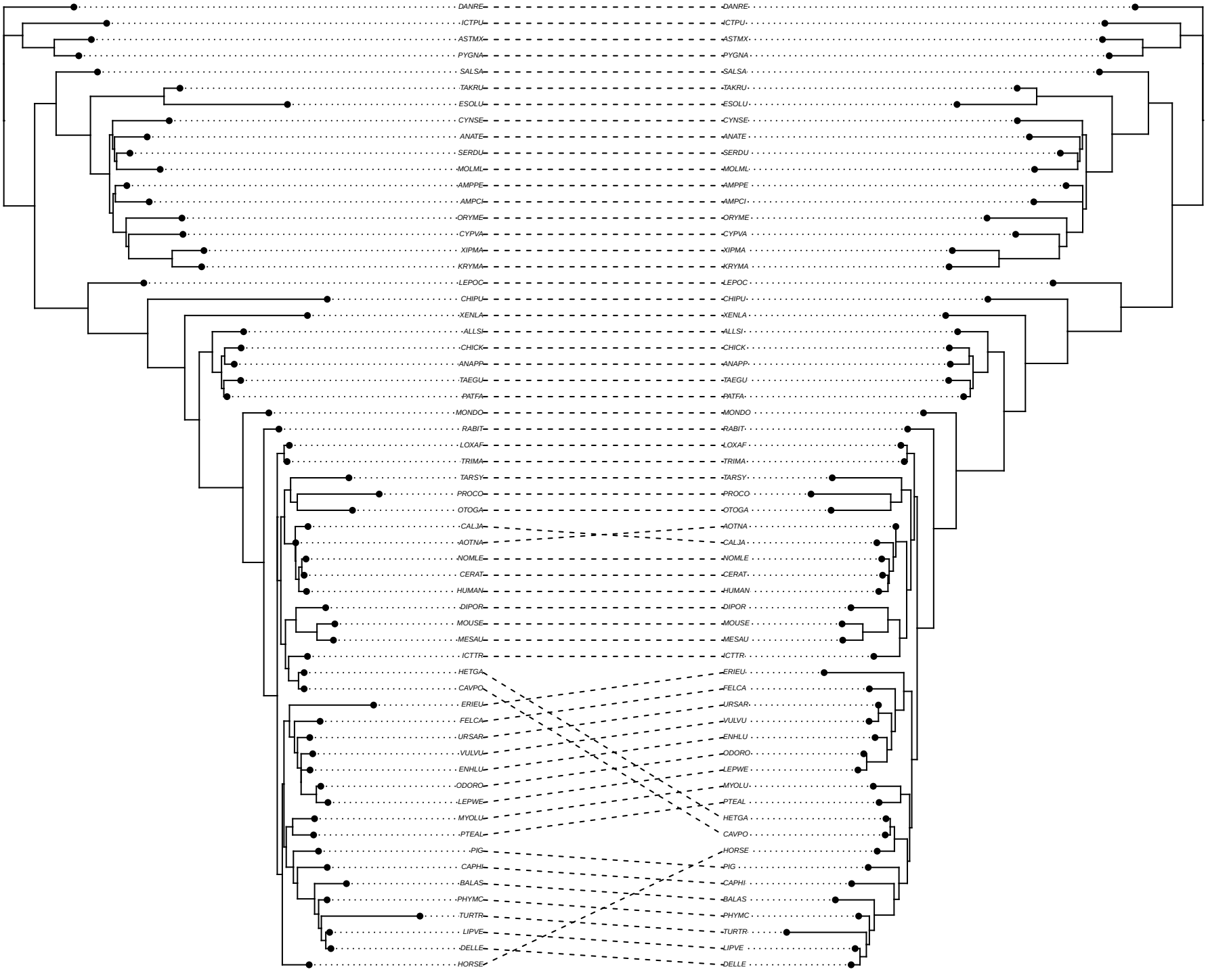
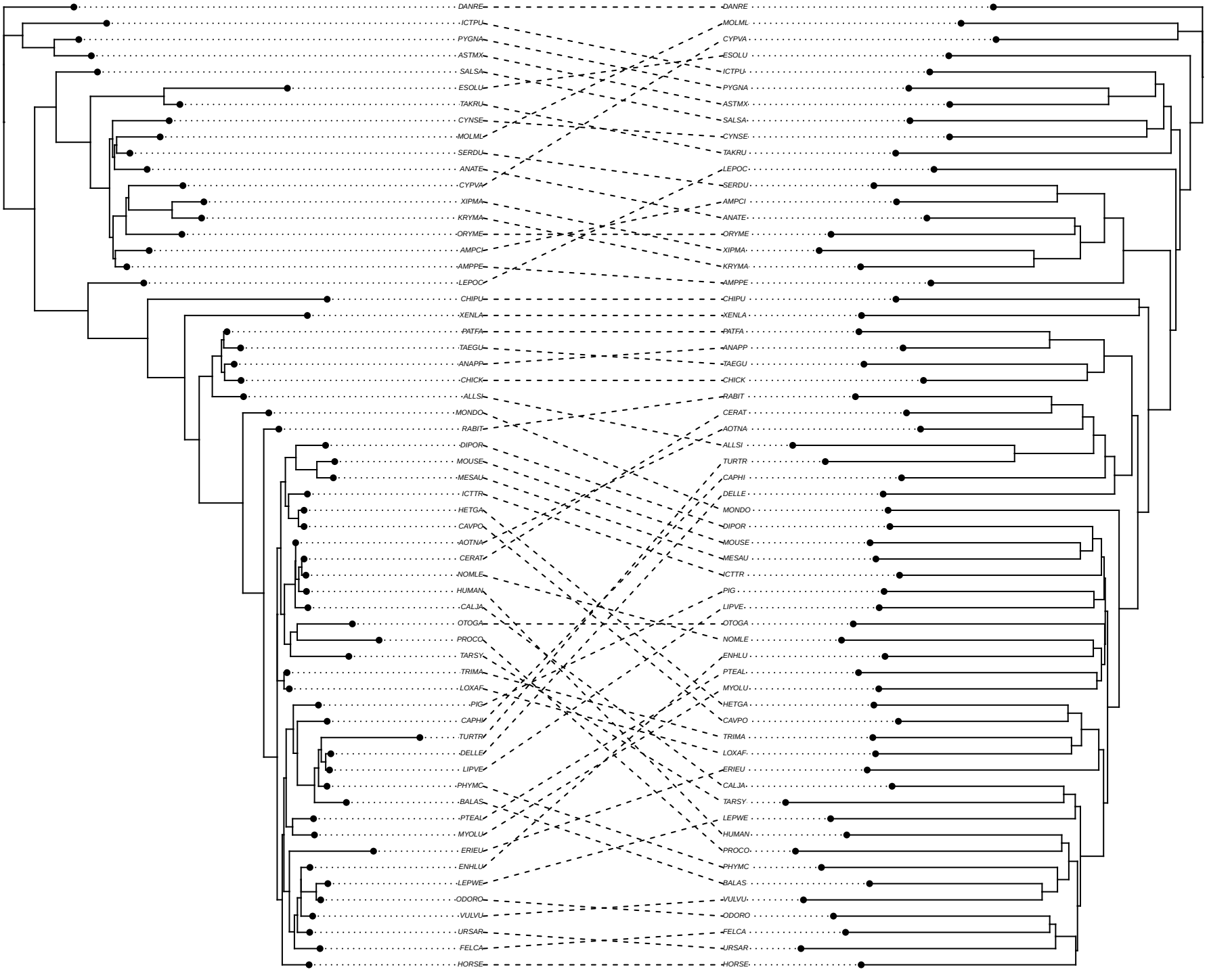

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
CH−astral
CH−ncbi−astral
CH−ncbi
clade
Teleostei
Shark + gar
Amphibia
Sauropsida
Laurasiatheria
Euarchontoglires
Afrotheria + opossum
Outgroup (DANRE)
Other

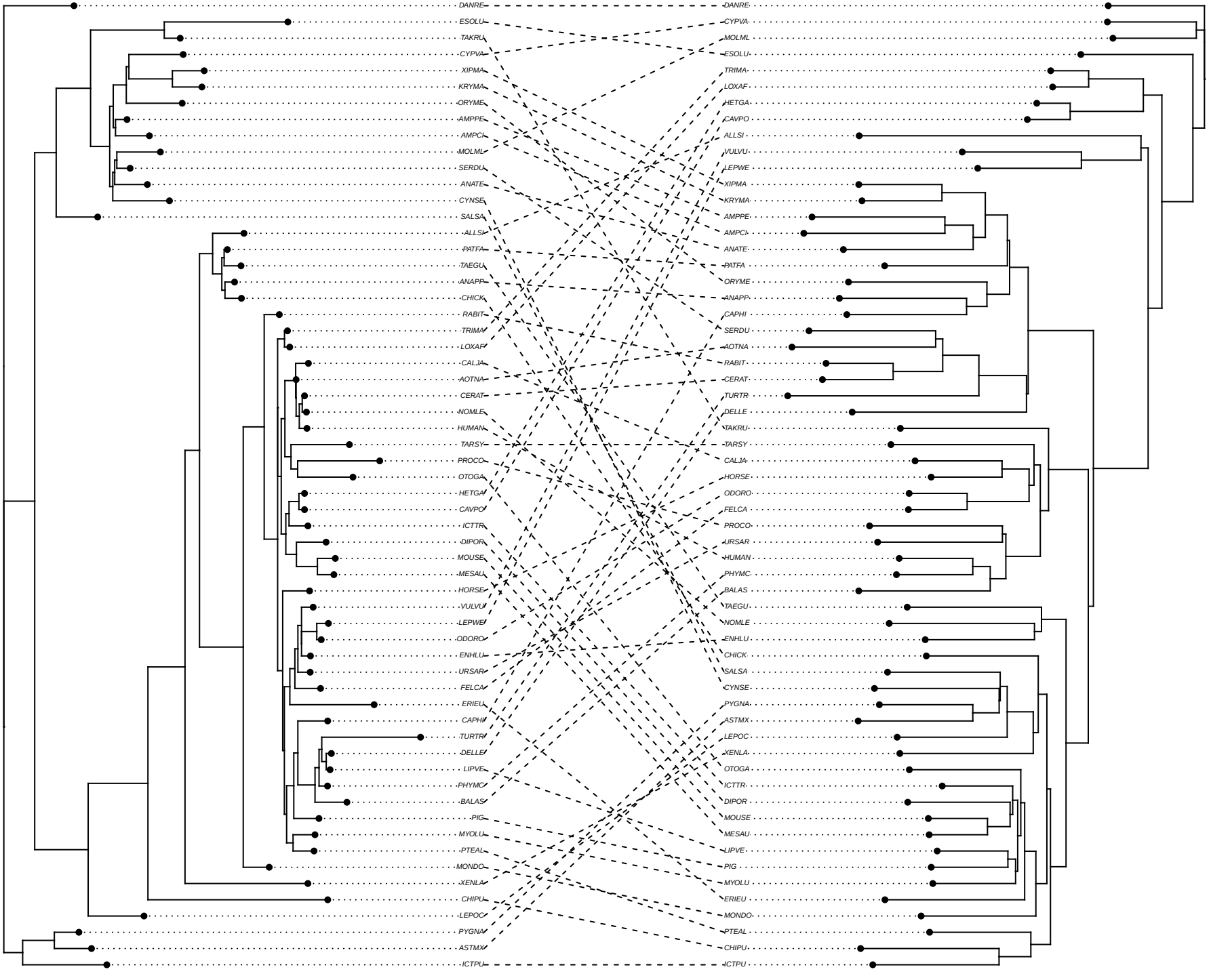

IQ−TREE vs Phylolformer
LOXAF
TRIMA
CAVPO
HETGA
ICTTR
MESAU
MOUSE
DIPOR
TARSY
PROCO
OTOGA
CALJA
HUMAN
NOMLE
CERAT
AOTNA
HORSE
ERIEU
VULVU
ENHLU
LEPWE
ODORO
URSAR
FELCA
PTEAL
MYOLU
BALAS
PHYMC
LIPVE
DELLE
TURTR
CAPHI
PIG
RABIT
MONDO
PATFA
TAEGU
ANAPP
CHICK
ALLSI
XENLA
CHIPU
LEPOC
ANATE
SERDU
MOLML
CYNSE
AMPCI
AMPPE
ORYME
XIPMA
KRYMA
CYPVA
TAKRU
ESOLU
SALSA
ICTPU
ASTMX
PYGNA
LOXAF
TRIMA
CAVPO
HETGA
ERIEU
ICTTR
VULVU
LEPWE
NOMLE
ENHLU
TARSY
CALJA
HORSE
ODORO
FELCA
PROCO
HUMAN
BALAS
PHYMC
URSAR
OTOGA
PTEAL
MYOLU
MESAU
MOUSE
AOTNA
DIPOR
SERDU
CERAT
RABIT
LIPVE
PIG
MONDO
DELLE
CAPHI
ANAPP
PATFA
ANATE
AMPCI
AMPPE
XIPMA
KRYMA
ORYME
TAEGU
CHICK
CHIPU
TURTR
XENLA
LEPOC
TAKRU
CYNSE
SALSA
ICTPU
ASTMX
PYGNA
ALLSI
ESOLU
MOLML
CYPVA

IQ-TREE vs CMAPLE

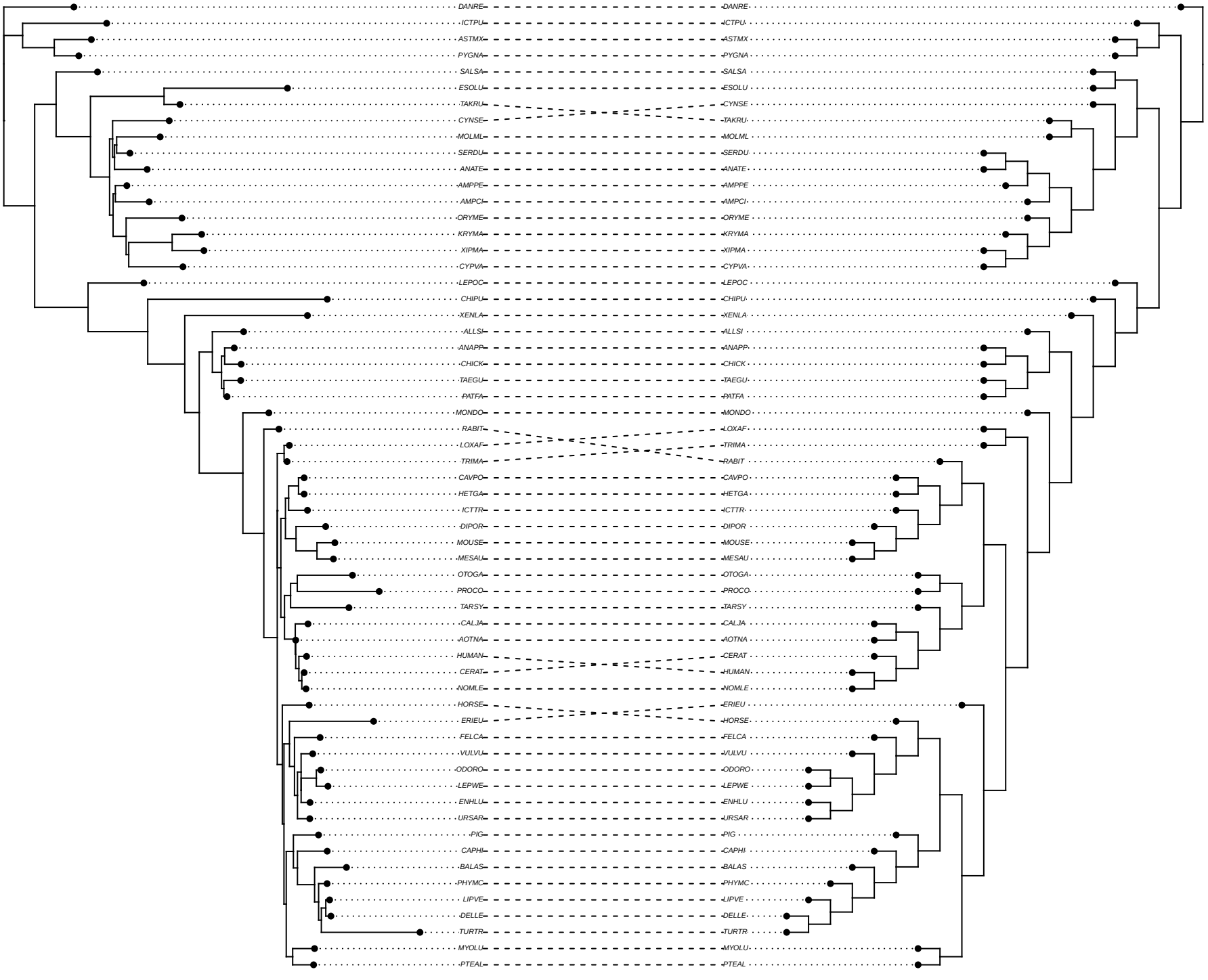



IQ−TREE vs MSA−Pattention
DANRE
ICTPU
PYGNA
ASTMX
SALSA
ESOLU
TAKRU
CYNSE
MOLML
SERDU
ANATE
CYPVA
XIPMA
KRYMA
ORYME
AMPCI
AMPPE
LEPOC
CHIPU
XENLA
PATFA
TAEGU
ANAPP
CHICK
ALLSI
MONDO
RABIT
DIPOR
MOUSE
MESAU
ICTTR
HETGA
CAVPO
AOTNA
CERAT
NOMLE
HUMAN
CALJA
OTOGA
PROCO
TARSY
TRIMA
LOXAF
PIG
CAPHI
TURTR
DELLE
LIPVE
PHYMC
BALAS
PTEAL
MYOLU
ERIEU
ENHLU
LEPWE
ODORO
VULVU
URSAR
FELCA
HORSE
DANRE
MOLML
CYPVA
ESOLU
ICTPU
PYGNA
ASTMX
SALSA
CYNSE
TAKRU
LEPOC
SERDU
AMPCI
ANATE
ORYME
XIPMA
KRYMA
AMPPE
CHIPU
XENLA
PATFA
ANAPP
TAEGU
CHICK
RABIT
CERAT
AOTNA
ALLSI
TURTR
CAPHI
DELLE
MONDO
DIPOR
MOUSE
MESAU
ICTTR
PIG
LIPVE
OTOGA
NOMLE
ENHLU
PTEAL
MYOLU
HETGA
CAVPO
TRIMA
LOXAF
ERIEU
CALJA
TARSY
LEPWE
HUMAN
PROCO
PHYMC
BALAS
VULVU
ODORO
FELCA
URSAR
HORSE

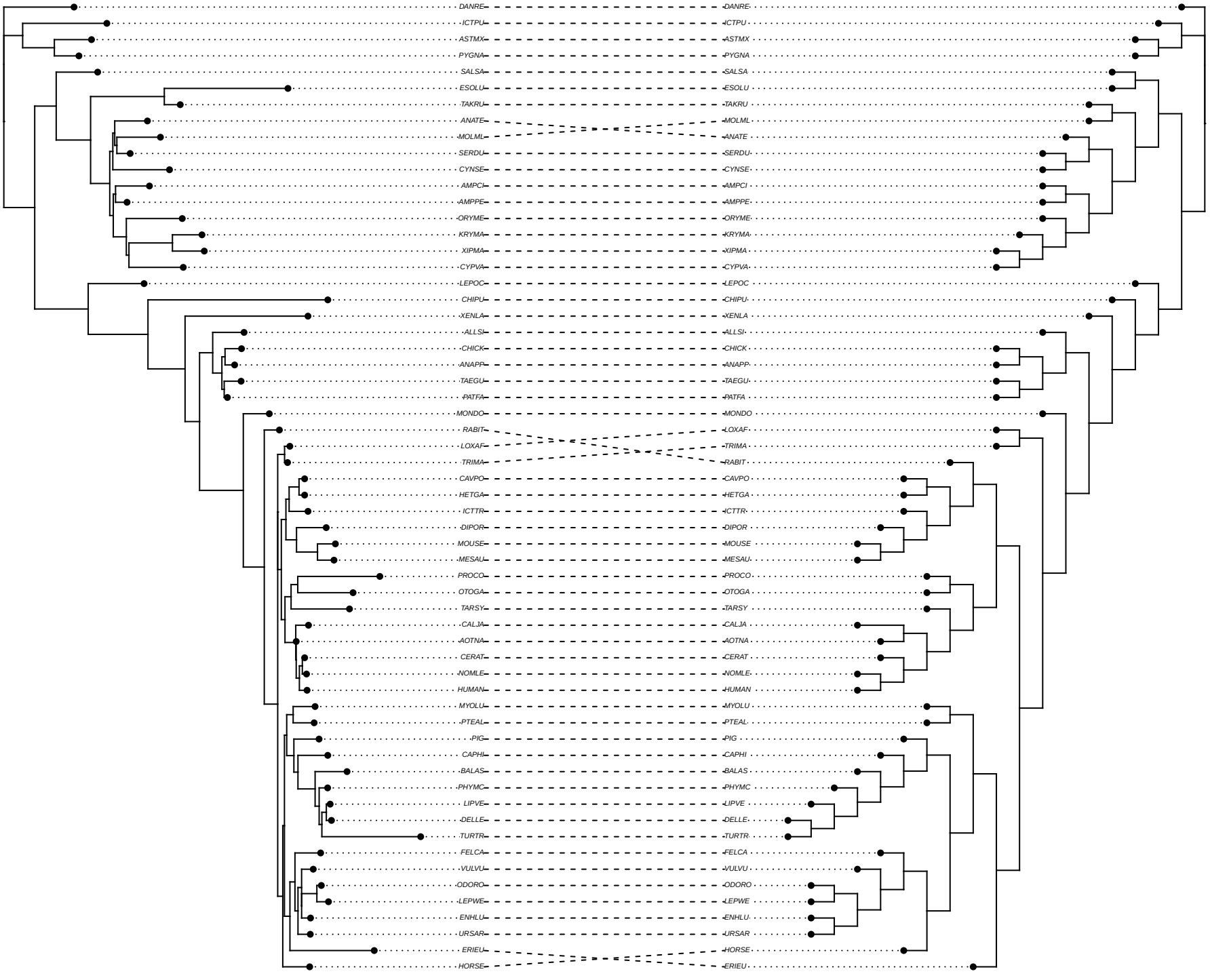
IQ−TREE vs MSA−T embedding



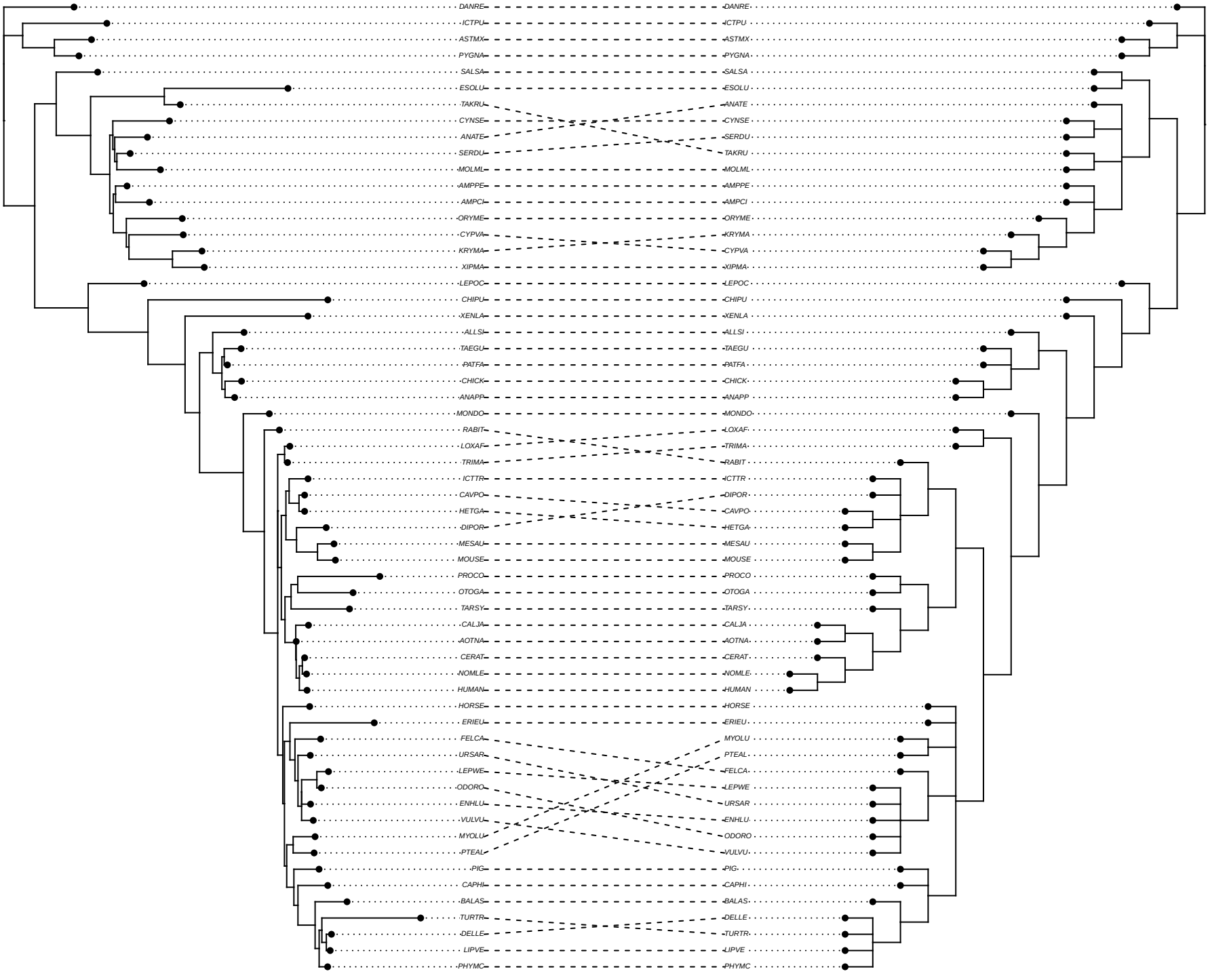
IQ−TREE vs CH−astral

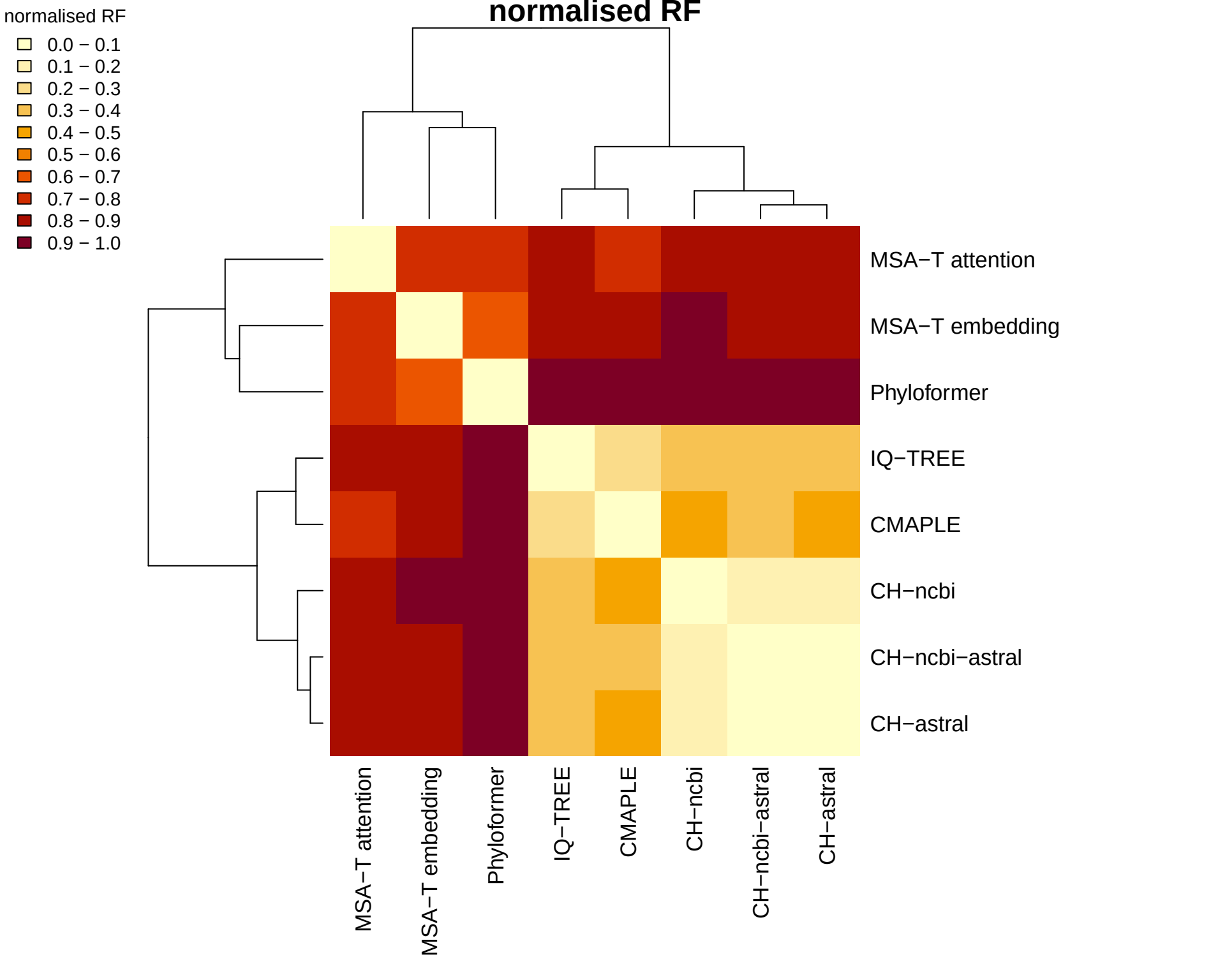


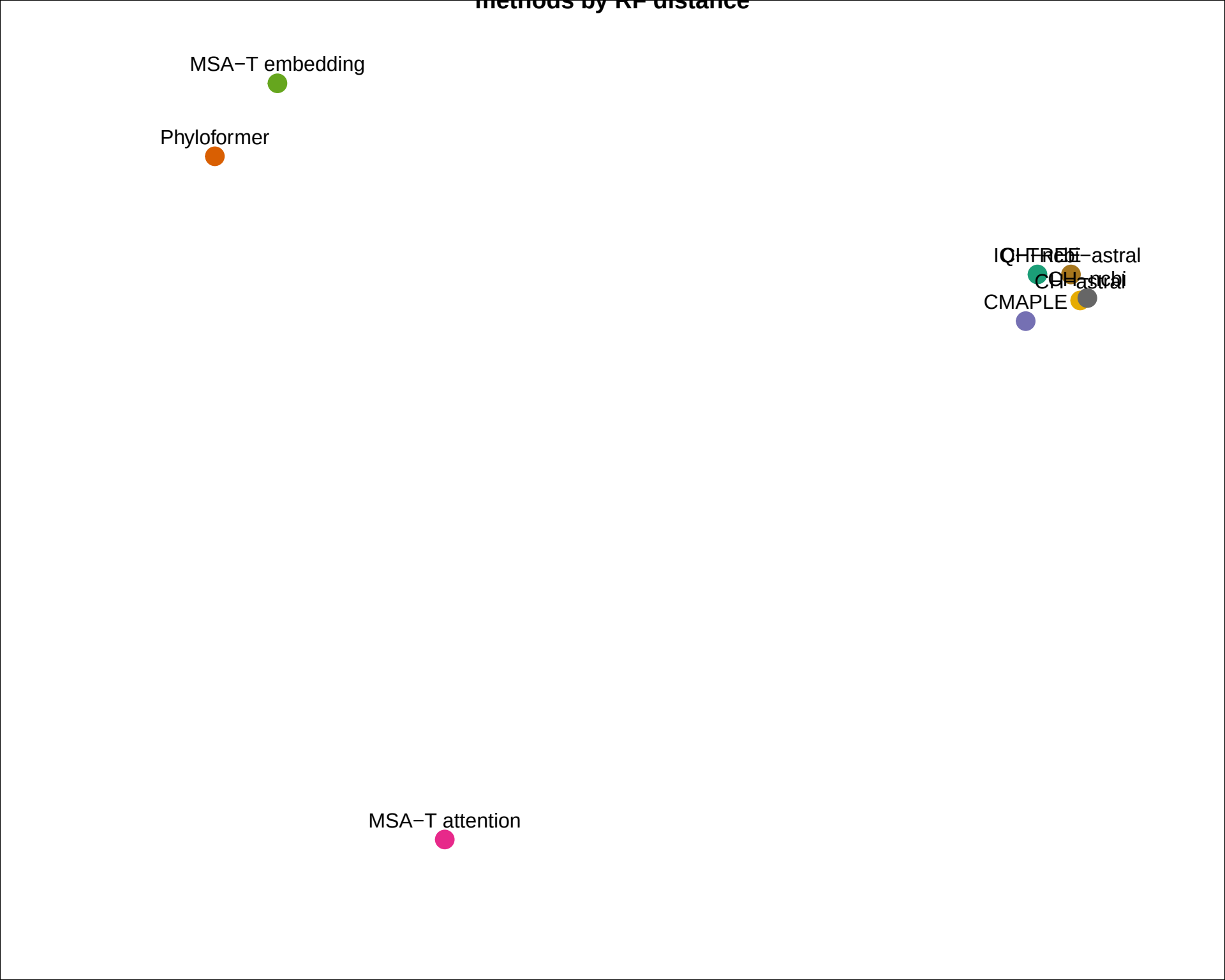
IQ-TREE vs CH-ncbi-astral

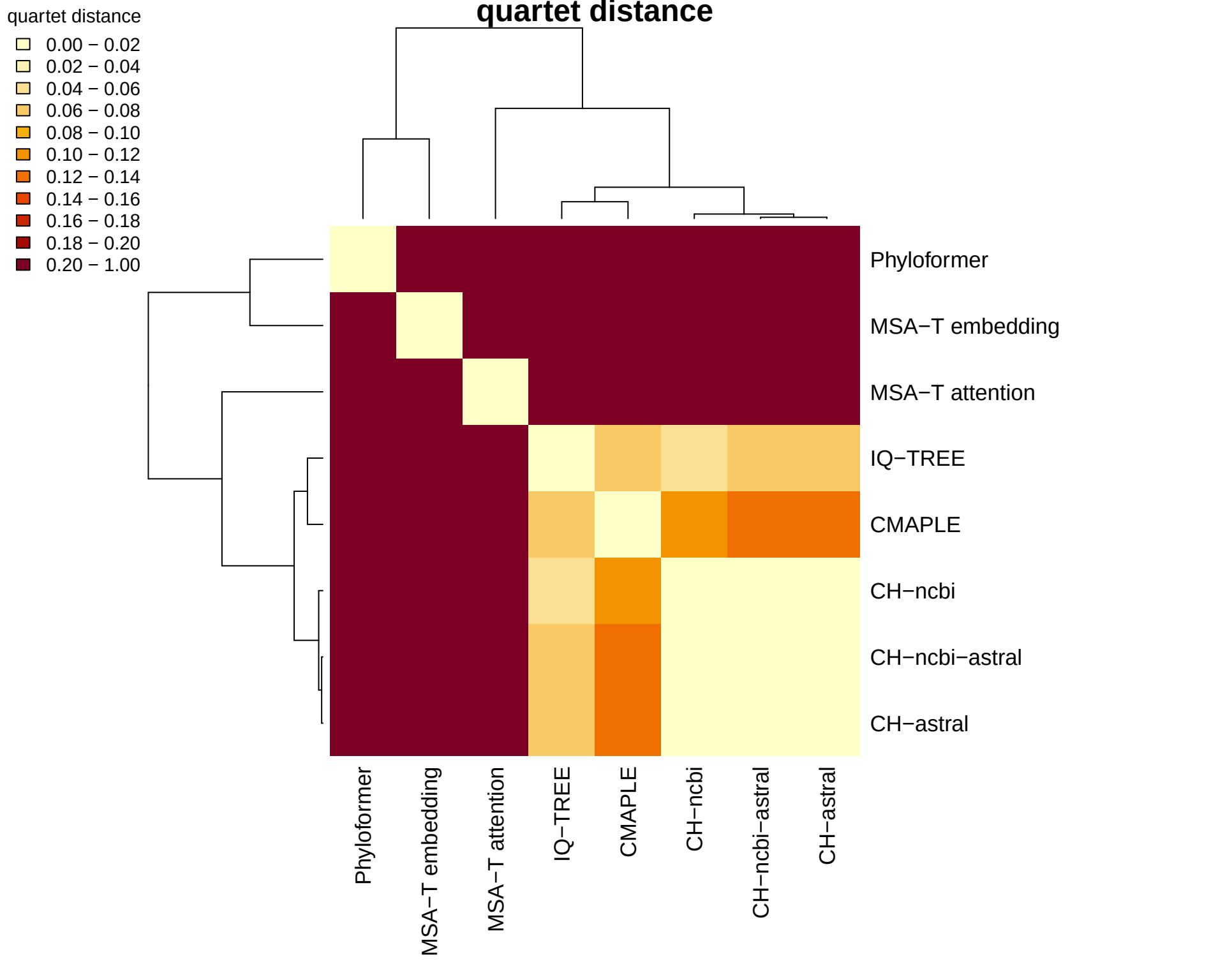


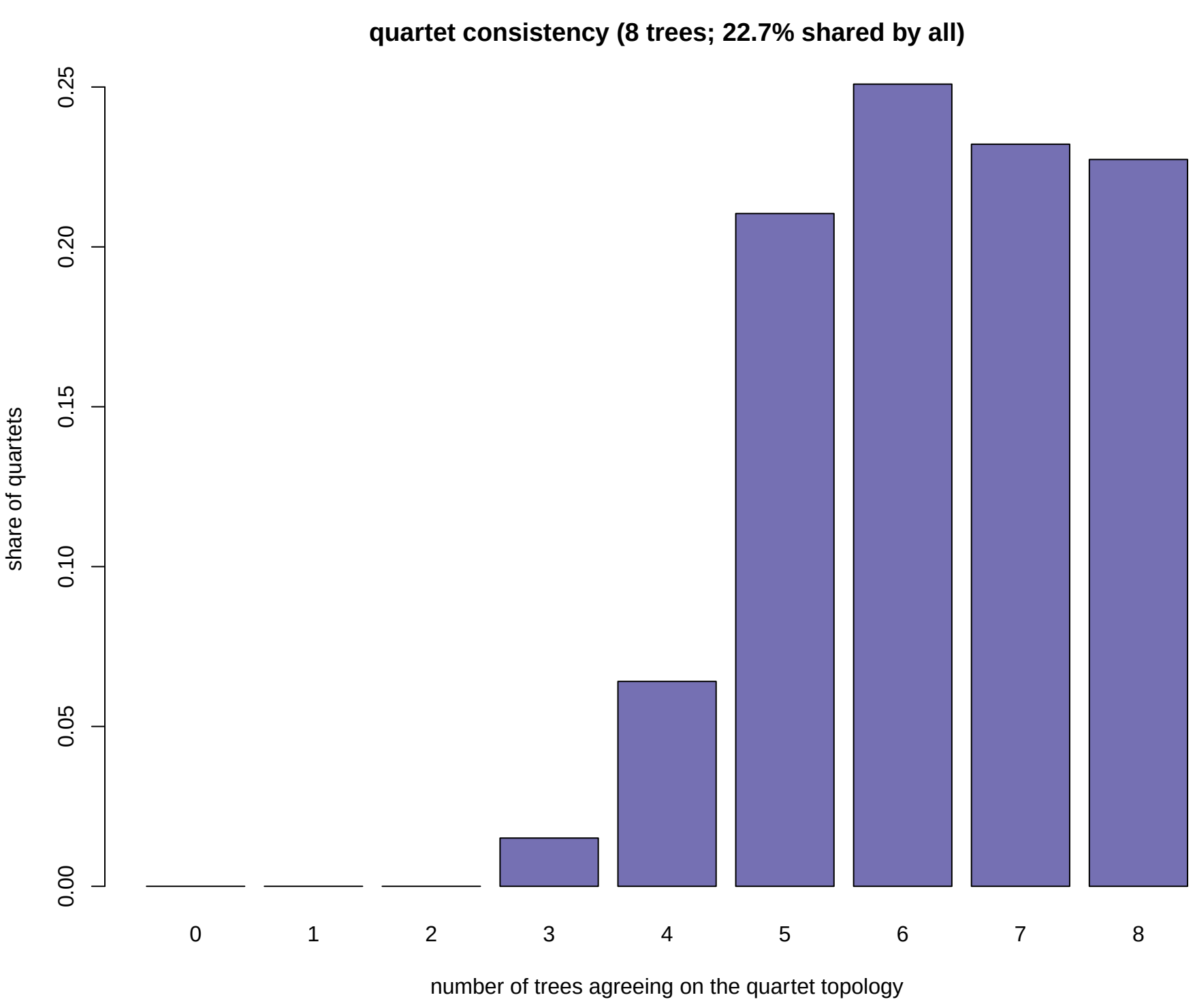
IQ-TREE vs CH-ncbi

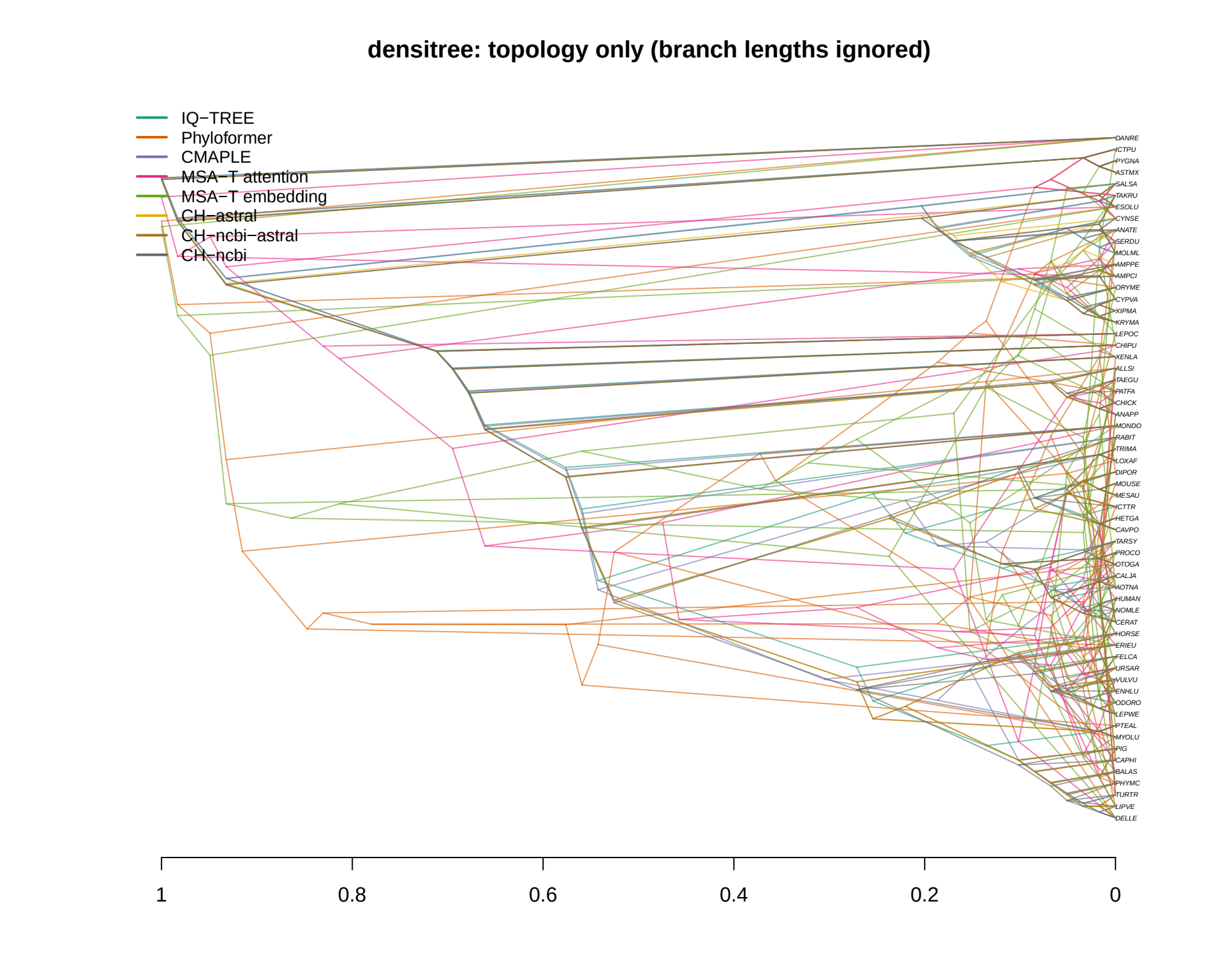


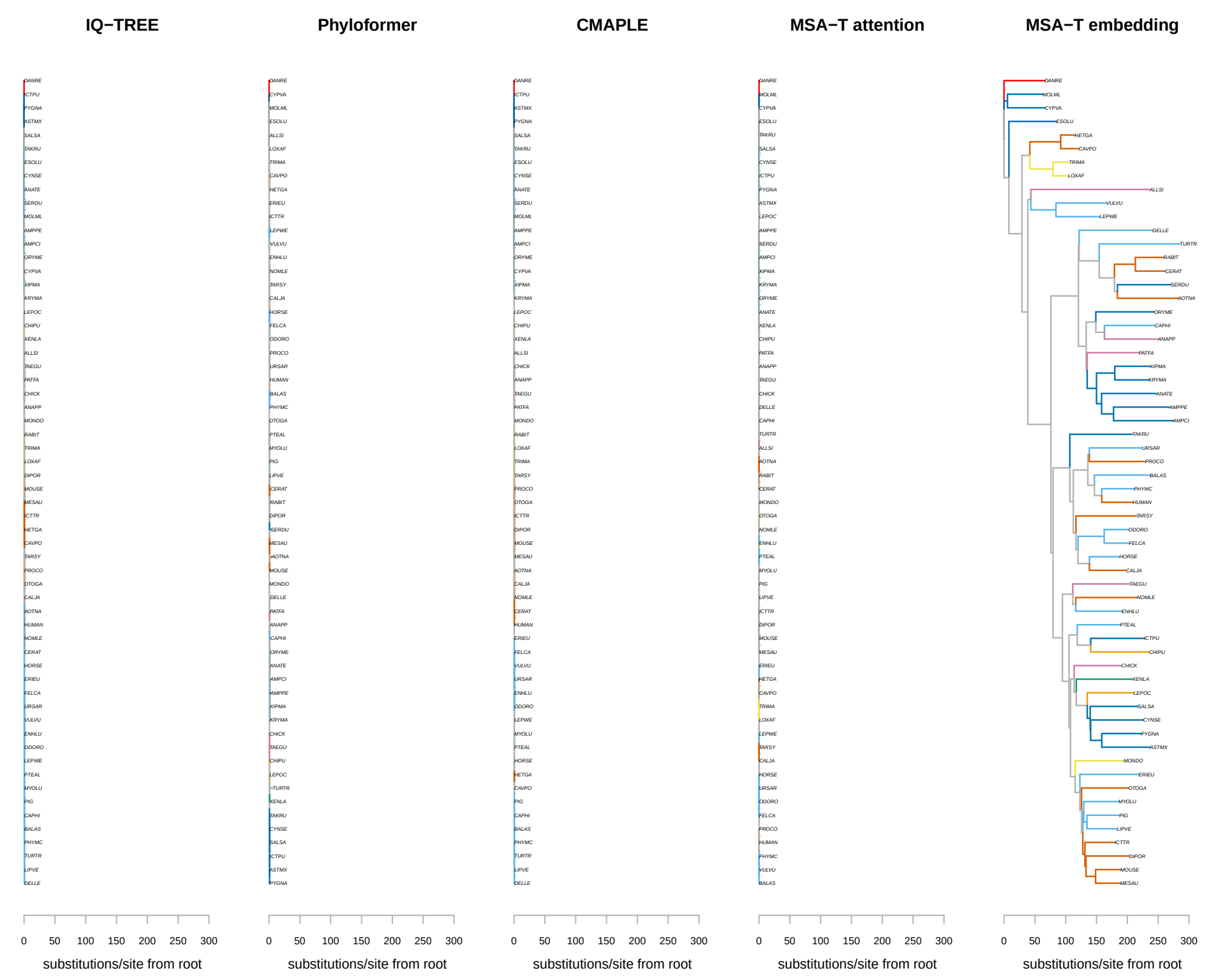

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T attention
MSA−T embedding
Phyloformer
IQ−TREE
CMAPLE
CH−ncbi
CH−ncbi−astral
CH−astral
MSA−T attention
MSA−T embedding
Phyloformer
IQ−TREE
CMAPLE
CH−ncbi
CH−ncbi−astral
CH−astral


methods by RF distance
MSA−T embedding
Phyloformer
IQ−TREE
FastTree
CH−astral
CH−nrbi
CMAPLE
MSA−T attention

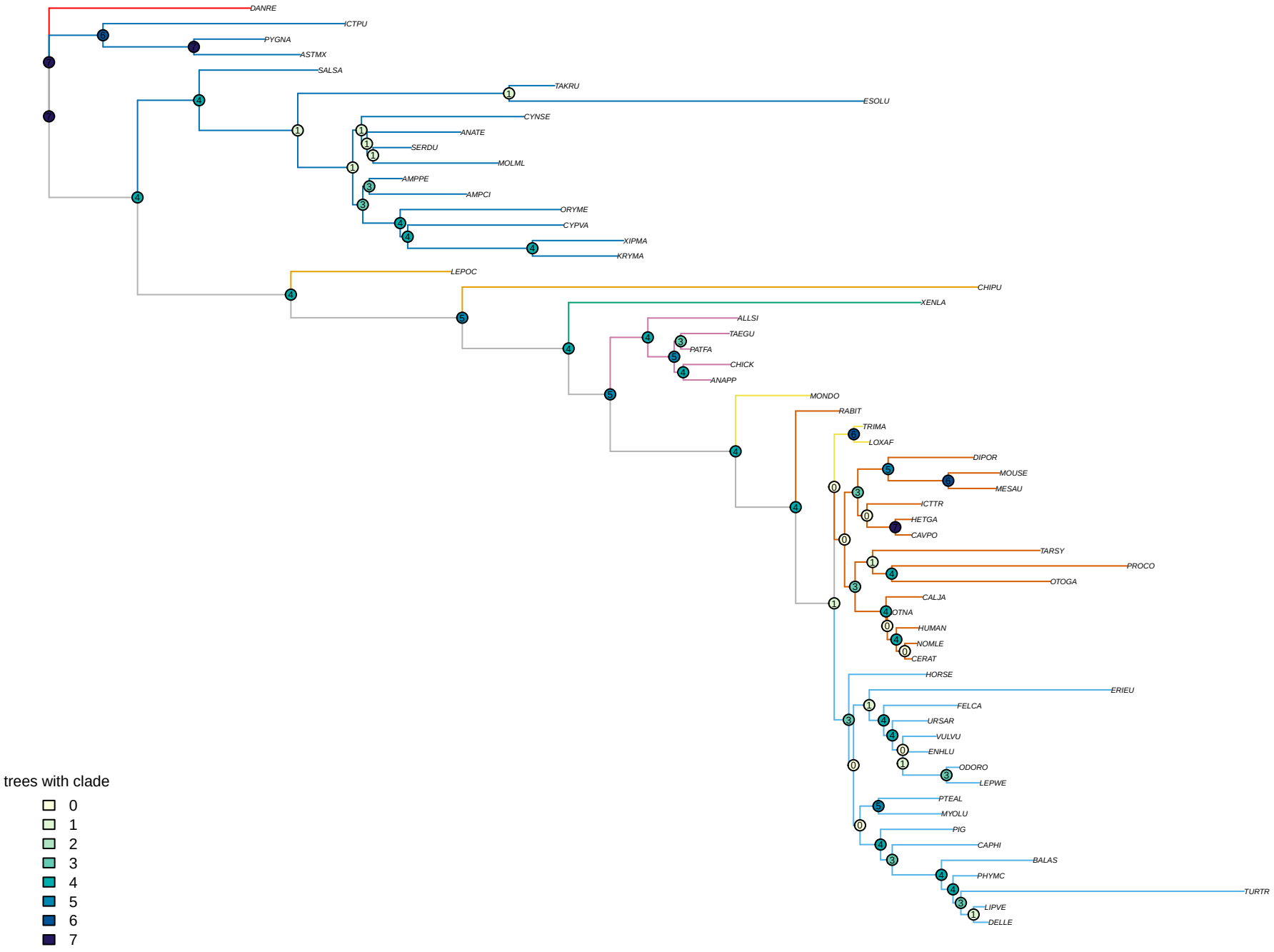

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
Phyloformer
MSA−T embedding
MSA−T attention
IQ−TREE
CMAPLE
CH−ncbi
CH−ncbi−astral
CH−astral
Phyloformer
MSA−T embedding
MSA−T attention
IQ−TREE
CMAPLE
CH−ncbi
CH−ncbi−astral
CH−astral


quartet consistency (8 trees; 22.7% shared by all)
share of quartets
0.00
0.05
0.10
0.15
0.20
0.25
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

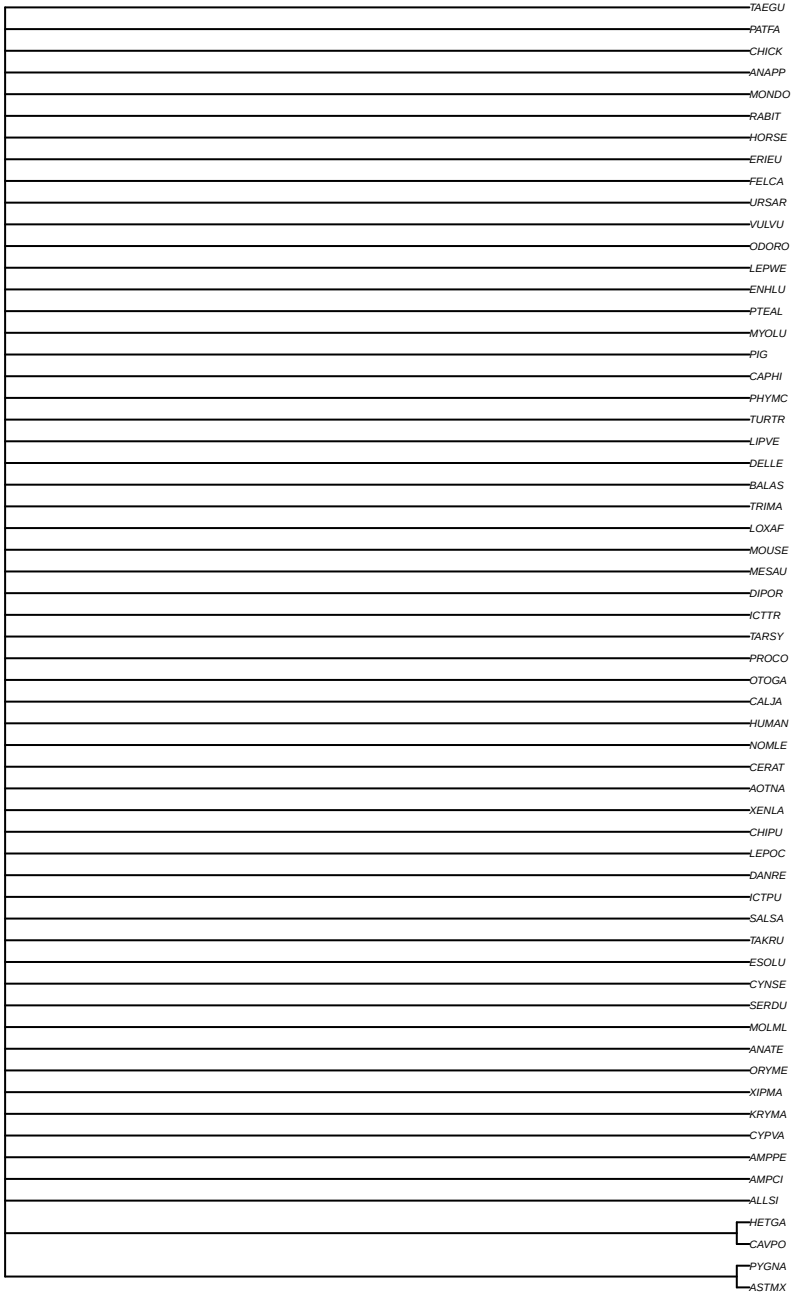

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
CH−astral
CH−ncbi−astral
CH−ncbi
DANRE
ICTPU
PYGNA
ASTMX
SALSA
TAKRU
ESOLU
CYNSE
ANATE
SERDU
MOLML
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CYPVA
XIPMA
KRYMA
LEPOC
CHIPU
XENLA
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LOXAF
DIPOR
MOUSE
MESAU
ICTTR
HETGA
CAVPO
TARSY
PROCO
OTOGA
CALJA
AOTNA
HUMAN
NOMLE
CERAT
HORSE
ERIEU
FELCA
URSAR
VULVU
ENHLU
ODORO
LEPWE
PTEAL
MYOLU
PIG
CAPHI
BALAS
PHYMC
TURTR
LIPVE
DELLE
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
DANRE
CTPU
PYGNA
ASTMX
SALSA
TAKRU
ESOLU
CYNSE
ANATE
SERDU
MOLML
AMPPE
AMPCI
ORYME
CYPVA
XIPMA
KRYMA
LEPOC
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HORSE
ERIEU
FELCA
URSAR
VULVU
ENHLU
ODORO
LEPWE
PTEAL
MYOLU
PIG
CAPHI
BALAS
PHYMC
TURTR
LIPVE
DELLE
0
50
100
150
200
250
300
substitutions/site from root

IQ−TREE clades: number of the other 7 trees that contain each clade



strict



majority rule

